A Defect Detection and Closed-Loop Control System with MCGI-YOLO for Circular Knitting Machines

Kangzhao Tan, Jiarui Zhang, Junjie Yu, Heng Wu, Jing Lin*, Diyu Li*, and Chunhua He*

***Abstract*—Defect detection is critical for circular knitting production, where persistent needle or yarn abnormalities lead to repeated fabric defects, resulting in severe material waste and economic losses. Existing methods are rarely tailored to circular knitting machines and tend to be cost-prohibitive, limiting large-scale industrial deployment. To address this issue, we present a low-cost, dedicated defect detection and closed-loop control system for circular knitting machines, which triggers image acquisition from machine rotation, halts the machine upon defect detection to interrupt defect propagation, and alerts the operator for corrective intervention, forming a complete production–detection–control loop. To further improve detection accuracy for diverse fabric defect morphologies and scales, we propose a lightweight detector termed MCGI-YOLO. To strengthen boundary evidence against repetitive fabric texture, selective gradient residual enhancement (SGRE) learns to select and mix fixed gradient responses after each bottleneck in the P3 and P4 backbone stages. To reconstruct fine spatial responses with limited computation, subpixel-adaptive residual upsampling (SARU) adds a low-resolution filtered correction with content-dependent weights for the four upsampled positions. On a real-world dataset collected from a circular-knitting production line, MCGI-YOLO achieves 97.4% mAP50 and 65.5% mAP50–95, improving on the YOLOv8n baseline by 6.9 and 7.1 percentage points, respectively. The model achieves a pure-inference throughput of 491 FPS on an RTX 3090 at FP32 precision and batch size 1. The proposed system provides a cost-effective solution for real-time defect detection in circular-knitting production.**

***Index Terms*—Circular knitting machine, edge computing, fabric defect detection, selective gradient residual enhancement (SGRE), subpixel-adaptive residual upsampling (SARU), YOLO.**

I. Introduction

CIRCULAR knitting machines are key production equipment for high-volume weft-knit fabric, which is widely used in everyday clothing such as T-shirts and sportswear [1]–[3]. The International Textile Manufacturers Federation reported approximately 24,500 global shipments of circular knitting machines in 2025 [4]. Globally, approximately 92 million tonnes of textile waste are generated annually, highlighting the importance of reducing avoidable material losses [5]. Conventional quality checks are often performed after knitting is completed. A persistent needle or yarn abnormality can therefore generate repeated defects along the fabric before discovery, accumulating substantial fabric waste and economic losses [6]–[8]. To prevent such losses, real-time detection during knitting is needed to identify these defects early and enable timely intervention before further defective fabric is produced.

Edge computing offers a practical pathway toward such real-time detection, and has been increasingly adopted for industrial defect detection. In textile applications, Machado et al. deployed YOLO11n on a Jetson Orin Nano to detect solid-color cotton and linen fabrics on a fabric rolling line [9]. Yan et al. deployed an improved YOLOv5s model on a Huawei Ascend 310-based Atlas platform to detect warp-knitted lace on warp-knitting machines [10]. Beyond textile detection, Wang et al. and Zhang et al. deployed an improved YOLOv5s model for rail surface defect detection and an improved Faster R-CNN detector with convolutional feature reuse for UAV-based insulator defect detection, respectively, both on Jetson Xavier NX [11], [12]. Li et al. deployed LiteYOLO-ID on a Jetson TX2 NX for insulator defect detection [13], and Zhao et al. implemented KAM-Net on a Xilinx ZCU102 FPGA for glass defect segmentation [14]. These studies collectively demonstrate the applicability of edge computing to industrial defect detection in their respective scenarios. However, such platforms are not tailored to the sensing, control, and installation constraints of circular knitting machines. The relatively high cost of general-purpose computing platforms, together with the additional integration effort required for image acquisition and machine control, hinders economically viable deployment across large numbers of machines.

In terms of detection algorithms, YOLO-based detectors have been widely explored for industrial defect detection because of their efficient inference and explicit defect localization [15]. FA-YOLO combines receptive-field enhancement, attention, and adaptive spatial feature fusion [16]. DGHR-YOLO integrates deformable convolution for morphology adaptation with a high-level screening feature pyramid and lightweight feature processing [17]. Zhao et al. proposed attention-based multiscale feature fusion (AMFF), combining intralayer feature enhancement and cross-layer attention to strengthen weak defect representations on fabric and aluminum surfaces [18]. An and Wan introduced JDCBL, combining contextual Transformer features with balanced contrastive learning to address scale variation and long-tailed defect categories in fabric detection [19]. Wang et al. developed RRNet, a lightweight rotated fabric defect detector that combines position-guided fusion and multiscale rotation-guided strip coupling to model elongated and tilted defect structures [20].

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Together, these methods advance the representation of subtle defects, irregular shapes, and scale variation. Related work on photovoltaic-cell and pipeline-weld detection also explores task-specific feature pyramids, contextual relations, and lightweight rotated representations [21]–[23]. However, these methods are tailored to their respective detection targets and may not be directly applicable to defect detection on circular knitting machines. Defects on circular knitting machines exhibit diverse morphologies and scales, ranging from horizontal and vertical elongated patterns to localized holes. In addition, modules such as deformable convolution and attention often introduce substantial computational overhead, making them less suitable for deployment on resource-constrained edge devices.

Motivated by these limitations, we design and implement a low-cost, dedicated defect detection and closed-loop control system for circular knitting machines. The system integrates a power adapter, a global-shutter camera with LED illumination, a coaxial serializer/deserializer (SerDes) image-transmission link, a Hall sensor, and an edge device. For defect detection within this system, we develop MCGI-YOLO, a lightweight detector that combines selective boundary enhancement with adaptive spatial reconstruction. SGRE combines fixed local gradients with learned selection and pointwise mixing to strengthen defect-boundary evidence while preserving the original features. SARU filters low-resolution features and predicts four subpixel-position weights to adapt the residual reconstruction before lateral fusion. Together, the components connect progressive boundary refinement with content-dependent spatial reconstruction within a compact parameter budget.

The main contributions of our work are as follows.

1) We design and implement a dedicated defect detection and closed-loop system for circular knitting machines, integrating image acquisition, camera transmission, edge inference, and local machine-stop control, and demonstrate its deployment on a real production line.
2) We develop SGRE to incorporate fixed gradient evidence through learned spatial selection and channel mixing. Its residual enhancement follows every bottleneck in the P3 and P4 backbone C2f blocks, preserving the original feature paths.
3) We design SARU to reconstruct fine spatial responses with low-resolution filtering and content-dependent subpixel weights. A shared positional weighting map modulates a residual correction to nearest-neighbor upsampling, keeping the learned convolutional cost linear in channel count.

The remainder of this paper is organized as follows. Section II presents the defect detection and closed-loop control system. Section III describes MCGI-YOLO and its two components. Section IV defines the dataset and evaluation protocol. Section V reports detector comparisons, component ablation, and model deployment. Section VI concludes.

II. Defect Detection and Closed-Loop Control System

A. System Overview

We design and implement a dedicated defect detection and closed-loop control system for circular knitting machines, as shown in Fig. 1. The closed-loop workflow begins when the operator starts the machine. The Hall sensor detects machine rotation and triggers the edge device to activate the camera and LED illumination for image acquisition and defect detection. When a defect is detected, the edge device immediately sends a stop command to the machine controller to prevent further defect propagation, uploads the defect image and record to the cloud, and pushes an alert notification to the operator. After the operator inspects the reported defect, resolves the production fault, and restarts the machine, the rotation signal reactivates detection. This sequence forms a complete production–detection–control loop. The system is organized into three functional subsystems: power supply, image acquisition and transmission, and processor. The following subsections describe their designs and hardware implementation.

B. Power Supply Subsystem

To accommodate different industrial supply voltages across countries and regions, we design a power adapter with a 100–500 V AC input range and a 24 V DC output rated at 3 A. The adapter first supplies this output to the device. The device then supplies the LED illumination. By adjusting the current delivered to the LEDs, the device controls the illumination brightness to suit the image-acquisition conditions. This arrangement integrates LED power delivery and brightness control into the device.

C. Image Acquisition and Transmission Subsystem

The rapid rotation of circular knitting machines requires imaging that preserves defect geometry and fine spatial detail. We therefore select a SmartSens SC130GS global-shutter sensor (SmartSens Technology Co., Ltd., Shanghai, China) at 1280×1024 resolution, which exposes all pixels simultaneously to avoid motion-induced distortion from sequential row exposure. At a camera-to-fabric distance of approximately 10 cm, the field of view is approximately 16×12 cm, with the 16 cm dimension aligned with the horizontal direction of fabric motion.

Factory machinery creates an electromagnetically noisy environment, and installation requires the camera and edge device to be separated by a cable run. We therefore use serializer/deserializer (SerDes) transmission over shielded coaxial cable to accommodate remote camera placement and the interference environment. Specifically, the THCV241A serializer and THCV242A deserializer (THine Electronics, Inc., Tokyo, Japan) bridge the camera-side and processor-side MIPI CSI-2 interfaces. The link supports a maximum serial line rate of 4.0 Gb/s per lane, corresponding to a rated effective rate of 3.2 Gb/s [24], [25].

The camera is configured to transmit 1280×1024 RAW10 images at 30 frames/s. With $b = 10$ bits per pixel, the active-image payload rate is

$$\begin{aligned} R_{\mathrm{img}} &= WHbf_{\mathrm{cam}} \\ &= 1280 \times 1024 \times 10 \times 30 \\ &= 393.216\ \mathrm{Mb/s} \end{aligned} \tag{1}$$

where W and H denote the image width and height, and f_{cam} is the camera acquisition rate. This corresponds to only 12.3%

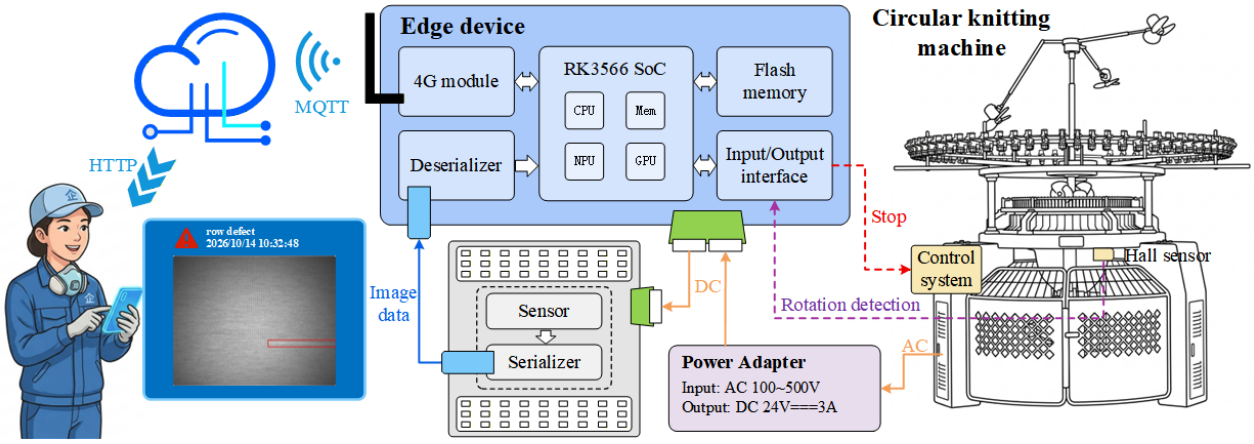


Fig. 1. The device comprises LED illumination control, a coaxial image link to the RK3566, 4G cloud connectivity, and I/O for rotation sensing and machine control.

of the rated single-lane effective capacity, providing a stable image input for the detection system.

D. Processor Subsystem

The processor is selected according to the detection rate required by fabric motion and the camera field of view. Taking a typical circular knitting machine as an example, with a machine diameter of 40 in ($D = 1.016$ m) and a rotation period of $T = 2$ s, the fabric travels approximately $\pi D = 3.192$ m per revolution. With an effective field-of-view width of $w = 0.16$ m along the direction of motion, the required frame rate is

$$f_{\min} = \frac{\pi D}{T w} = \frac{\pi \times 1.016}{2 \times 0.16} \approx 9.97 \text{ frames/s} \tag{2}$$

We set the effective detection-throughput requirement to 10 frames/s. For this throughput target, we select the Rockchip RK3566 (Rockchip Electronics Co., Ltd., Fuzhou, China) to balance processing capability, camera connectivity, and hardware cost. It integrates a quad-core Cortex-A55 CPU and an NPU rated at 1 TOPS for INT8 operations [26].

Table I summarizes the hardware specifications of platforms used in related defect detection studies. While high-performance platforms like the NVIDIA Jetson Orin Nano and Xilinx ZCU102 offer substantial computational power, their costs range from hundreds to over 3,000 USD, which can be prohibitive for large-scale deployment. In contrast, our device provides a cost-effective hardware basis for the proposed system at a hardware cost of 60 USD, offering a significant cost advantage over comparable industrial platforms and making it highly suitable for large-scale deployment in textile factories.

E. Hardware Implementation

Based on the design requirements of the subsystems described above, we develop the corresponding hardware shown in Fig. 2. The camera assembly (a) combines the image sensor and LED illumination. We design a camera-side interface board (b) that integrates the serializer and coaxial connector, and a mainboard (c) built around the RK3566. The mainboard integrates flash storage, the camera deserializer and coaxial connector, a 4G module with SIM-card and antenna interface,

TABLE I
HARDWARE PLATFORMS USED IN RELATED DEFECT DETECTION STUDIES.

Platform	Compute capability	Price (USD)
NVIDIA Jetson TX2 NX [13]	1.33 TFLOPS (FP16)	249
NVIDIA Jetson Xavier NX [11], [12]	21 TOPS (INT8)	399
NVIDIA Jetson Orin Nano [9]	40 TOPS (INT8)	499
Huawei Ascend 310 [10]	16 TOPS (INT8)	550
AMD Xilinx ZCU102 [14]	3.45 TOPS (INT8)†	3234
Our device	1 TOPS (INT8)	60

†AMD reference INT8 peak for three B4096 DPU cores.

input/output interface, and power-management circuitry. This board-level integration brings image reception, local inference, network communication, machine control, and illumination control into a dedicated edge device. Fig. 3 shows the system installed on a circular knitting machine in a real production line.

III. Detection Algorithm

A. Network Architecture

Fig. 4 shows the detector developed from YOLOv8n [27]. Selective gradient residual enhancement (SGRE) is inserted after every bottleneck in the P3 and P4 backbone C2f blocks, progressively incorporating selected boundary responses while preserving the original feature paths. Subpixel-adaptive residual upsampling (SARU) replaces the two top-down upsampling operations, using content-dependent weights for the four output positions of each low-resolution cell. The remaining backbone stages, bottom-up aggregation path, and three-scale detection head are retained [28], [29].

B. Selective Gradient Residual Enhancement

Knitting defects can produce narrow, weak boundaries against repetitive fabric texture. Their boundary responses provide complementary evidence to the features learned by convolution, but gradient magnitude alone also responds to normal yarn patterns. We therefore develop SGRE to select and mix gradient evidence before adding it to the original features

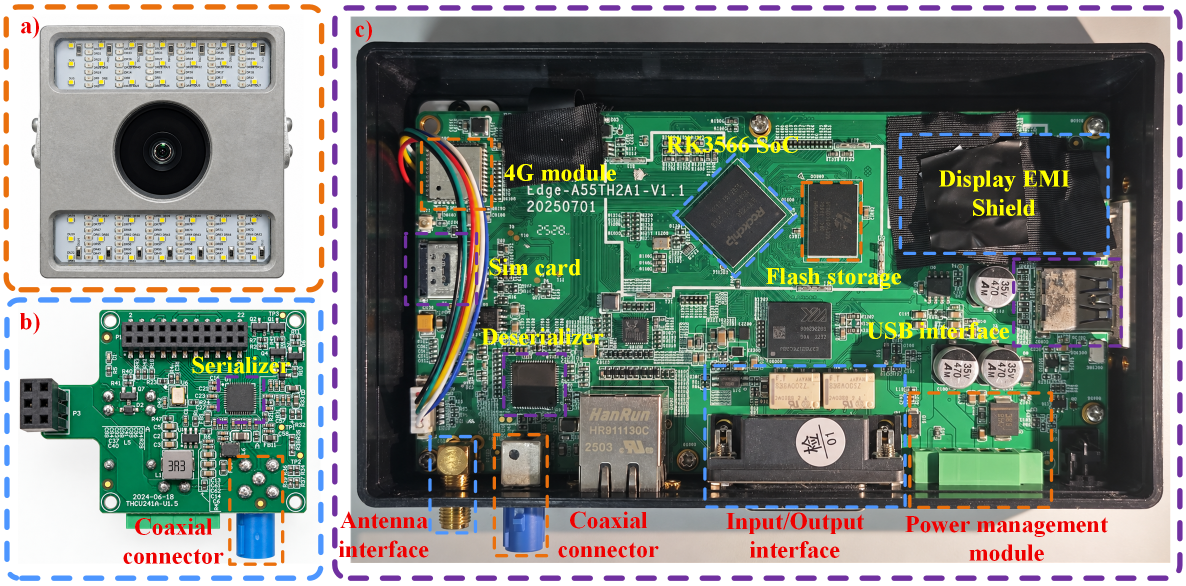


Fig. 2. Hardware of the defect detection system: (a) industrial camera with LED illumination; (b) camera-side interface board with a serializer and coaxial connector; and (c) RK3566 edge board with the 4G communication, flash storage, camera-interface, input/output interface, and power-management components.

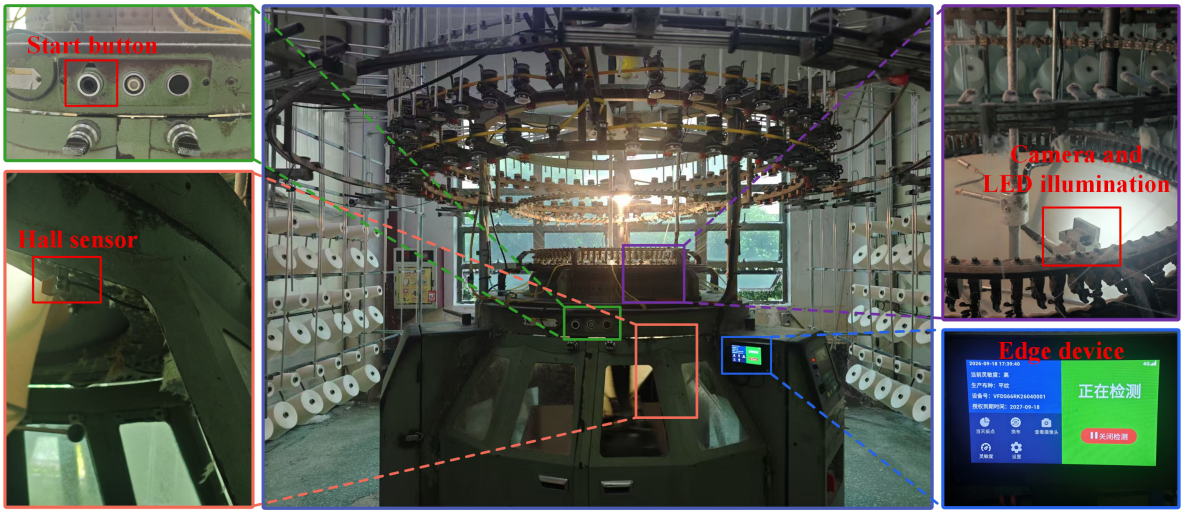


Fig. 3. Actual deployment on the circular knitting machine at Dongguan Minghai Textile Co., Ltd., Dongguan, Guangdong, China. The insets highlight the start button, Hall sensor, industrial camera, and RK3566 edge device, respectively.

[Fig. 5(a)]. The Sobel filters supply a fixed local derivative prior, whereas the selection and channel mixing are learned from the detection objective. SGRE is applied to all hidden channels after each complete bottleneck output, including its original shortcut, in the backbone P3 and P4 C2f blocks. This placement lets the enhanced output enter both the next bottleneck and the final C2f concatenation. P2 and P5 retain their original C2f blocks.

1) Fixed Gradient Extraction: For a bottleneck output $F \in \mathbb{R}^{B\times C\times H\times W}$, two depthwise convolutions compute gradients using normalized Sobel kernels:

$$K_x = \frac{1}{8}\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \qquad K_y = K_x^{\mathsf{T}} \tag{3}$$

The gradient evidence is formed as

$$E = |K_x *_{\mathrm{dw}} F| + |K_y *_{\mathrm{dw}} F| \tag{4}$$

where $*_{\mathrm{dw}}$ denotes channel-independent convolution with padding of one. Both absolute-value operations are performed before summation. Thus E has the same $B\times C\times H\times W$ shape as F. The two Sobel kernels remain fixed during training.

2) Selective Residual Enhancement: A learned depthwise convolution and Hardsigmoid predict channel-specific spatial selection weights from the original feature, while a separate pointwise convolution mixes the gradient evidence:

$$G = \mathrm{HS}(\mathrm{DWConv}_{3\times 3}(F)) \tag{5}$$

$$R = \mathrm{Conv}_{1\times 1}(E) \tag{6}$$

$$Z = F + G \odot R \tag{7}$$

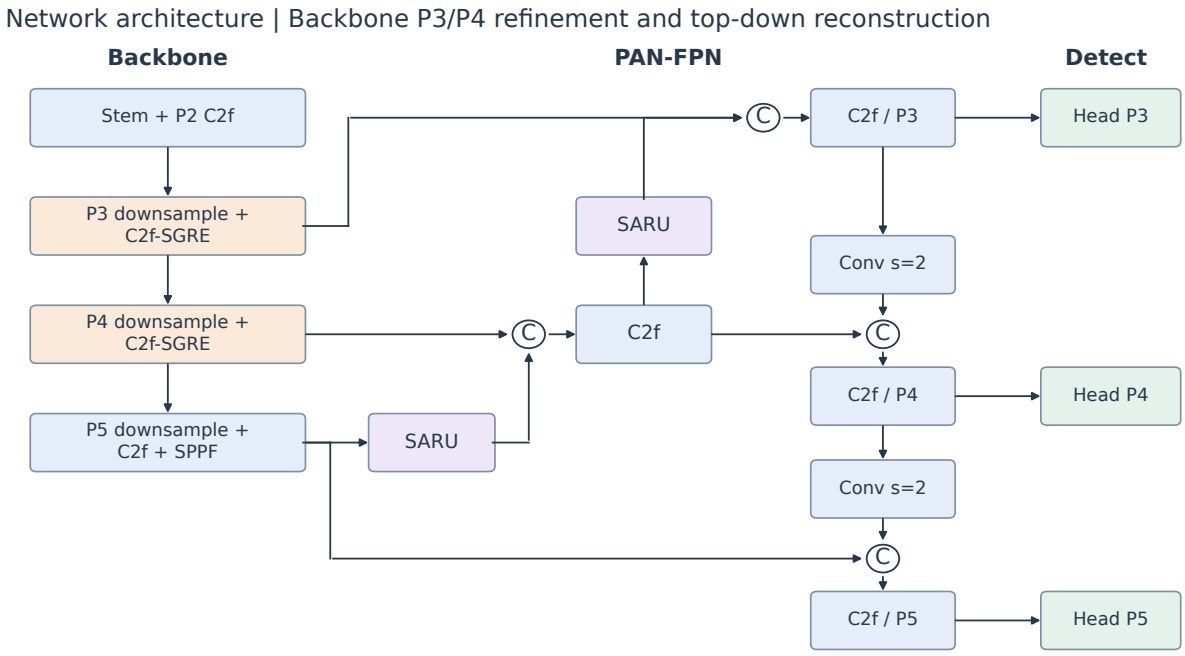


Fig. 4. Network architecture of the proposed detector. Each bottleneck in the backbone P3 and P4 C2f blocks is followed by SGRE. SARU replaces the two top-down upsampling operations. C denotes channel concatenation; downsampling convolutions and the three-scale detection head retain the YOLOv8n structure.

where $G, R, Z \in \mathbb{R}^{B\times C\times H\times W}$. The depthwise selection convolution includes a bias, whereas the C-to-C pointwise projection is bias-free. The selection weights regulate the contribution of gradient evidence at each channel and spatial position, and the residual connection retains the original feature. There is no partial-channel split inside SGRE. The selection convolution and pointwise projection are initialized to zero, giving $G = 0.5$ and $Z = F$ at initialization.

For a selected C2f block, let U, V denote its two initial hidden branches. The enhanced progressive outputs satisfy

$$Z_0 = V \qquad Z_i = \mathrm{SGRE}_i(\mathrm{Bottleneck}_i(Z_{i-1})) \tag{8}$$

for $i = 1, \ldots, n$. The original C2f fusion concatenates U, V, and all enhanced outputs $Z_1, \ldots, Z_n$. Each SGRE has independent learned weights. Under the YOLOv8n scaling used here, P3 and P4 each contain two such bottlenecks, giving four SGRE units in total.

3) Computational Cost: For one SGRE with C hidden channels, the added parameter count and convolutional MAC count are

$$P_{\mathrm{SGRE}} = C^2 + 10C \tag{9}$$

$$\mathrm{MAC}_{\mathrm{SGRE}} = BHW(C^2 + 27C) \tag{10}$$

The C^2 parameters belong to the pointwise projection; the depthwise selector contributes $9C$ weights and C biases. The fixed Sobel filters are stored as buffers rather than model parameters. In convolutional computation, however, both filters still contribute $18C$ MACs per spatial location, in addition to the selector's $9C$ and the projection's C^2 MACs. Absolute values, Hardsigmoid, and element-wise arithmetic introduce additional linear cost in $BCHW$. The complete added cost is obtained by summing over the four inserted units.

C. Subpixel-Adaptive Residual Upsampling

In the top-down feature pyramid, a low-resolution cell is expanded into four output positions before fusion with lateral features. These positions can cover different parts of a narrow defect or a hole boundary. Nearest-neighbor interpolation assigns the same response to all four positions and does not allow their reconstruction to depend on local content. SARU retains this parameter-free base and adds a locally filtered residual with separate content-dependent weights for the four subpixel positions [Fig. 5(b)]. The learned filtering and weight prediction operate at the low resolution to limit computation. Unlike a four-channel feature expansion, the four predicted channels encode positional weights shared across feature channels.

1) Base and Residual Paths: For $X \in \mathbb{R}^{B\times C\times H\times W}$, the two feature paths are

$$N = \mathrm{Nearest}_2(X) \tag{11}$$

$$R = \mathrm{Nearest}_2(\mathrm{DWConv}_{3\times 3}(X)) \tag{12}$$

where $N, R \in \mathbb{R}^{B\times C\times 2H\times 2W}$. The residual convolution is depthwise, bias-free, and uses padding of one. It filters neighboring low-resolution responses before expansion, whereas N preserves the baseline nearest-neighbor feature.

2) Subpixel Weight Prediction: A biased pointwise convolution maps C channels to four position-dependent weights, which are rearranged by PixelShuffle:

$$A = \mathrm{HS}(\mathrm{Conv}_{1\times 1}^{C\to 4}(X)) \tag{13}$$

$$G = \mathrm{PixelShuffle}_2(A) \tag{14}$$

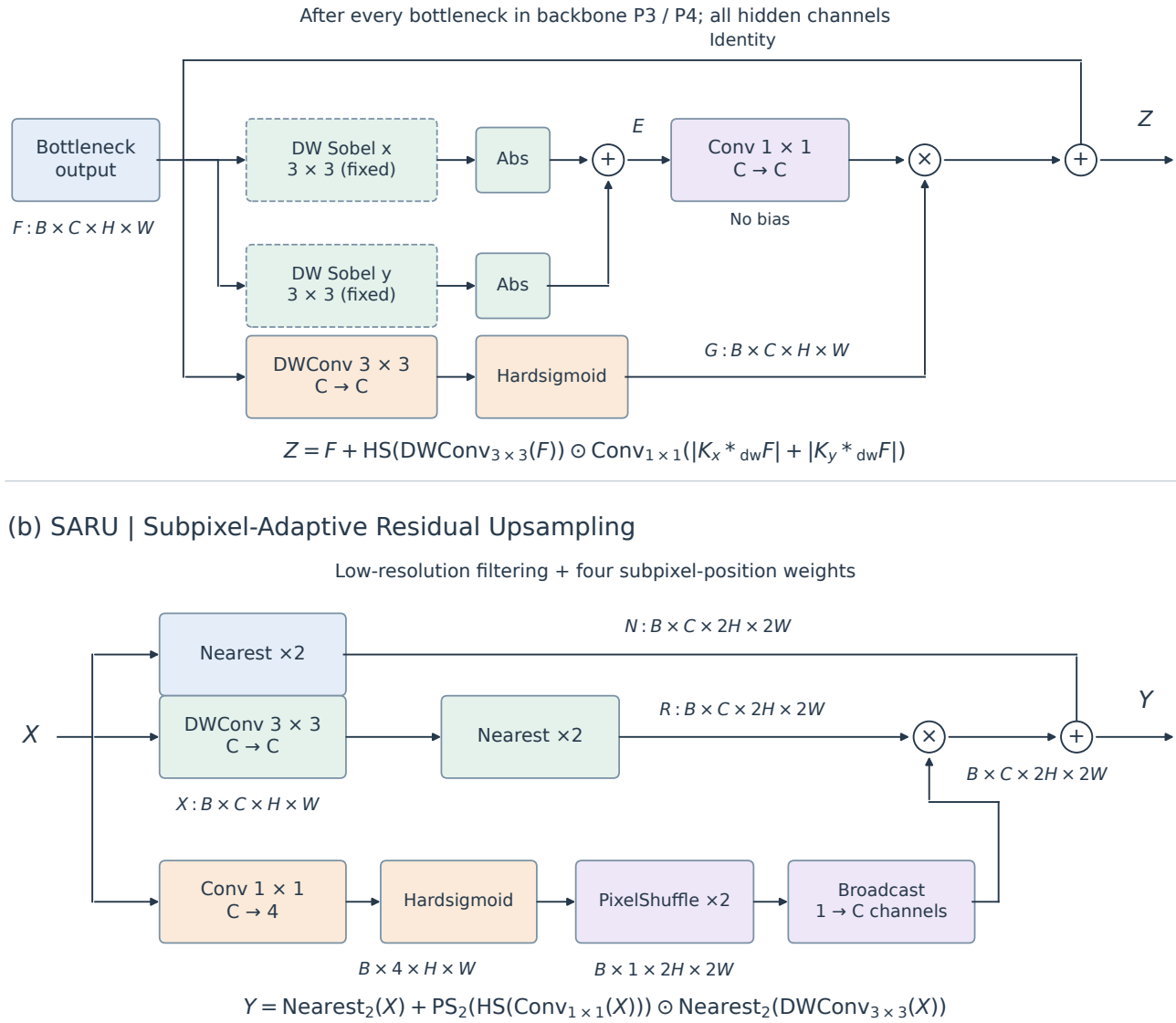


Fig. 5. Implementation-matched module structures: (a) SGRE applies fixed depthwise Sobel filters, channel-wise spatial selection, and learned pointwise mixing to a complete bottleneck output; (b) SARU adds a low-resolution filtered residual to nearest-neighbor upsampling. PixelShuffle rearranges four weights into four distinct subpixel positions; the resulting single-channel map is broadcast across feature channels before multiplication. Dashed blocks denote fixed filters.

Here, $A \in [0,1]^{B\times 4\times H\times W}$ and $G \in [0,1]^{B\times 1\times 2H\times 2W}$. For zero-based spatial indices h, w and offsets $i, j \in \{0,1\}$, the rearrangement is

$$G(b,1,2h+i,2w+j) = A(b,2i+j+1,h,w) \tag{15}$$

so the four predicted channels correspond to the top-left, top-right, bottom-left, and bottom-right positions. Before multiplication, G is broadcast along the channel dimension from $B \times 1 \times 2H \times 2W$ to $B \times C \times 2H \times 2W$ by reusing the same spatial weights for all C channels.

3) Residual Reconstruction and Cost: The reconstructed feature is

$$Y = N + G \odot R \tag{16}$$

where the multiplication uses channel broadcasting and $Y \in \mathbb{R}^{B\times C\times 2H\times 2W}$. This additive correction can increase or decrease a feature response because R is signed. Zero-initializing the depthwise residual convolution and the four-output predictor makes $R = 0$, $G = 0.5$, and $Y = N$ at initialization.

The added parameter and convolutional MAC counts are

$$P_{\mathrm{SARU}} = 13C + 4 \tag{17}$$

$$\mathrm{MAC}_{\mathrm{SARU}} = 13BCHW \tag{18}$$

The residual depthwise convolution contributes $9C$ weights, and the C-to-four predictor contributes $4C$ weights and four biases. Both convolutions operate at $H \times W$; PixelShuffle only rearranges weights, while nearest-neighbor interpolation copies values. The output multiplication and addition each operate on $4BCHW$ elements. Consequently, learned convolutional computation scales linearly with channel count without constructing a channel-by-neighborhood reassembly tensor. When

expressing convolutional computation as FLOPs, one MAC is counted as two operations; these analytic convolution counts are separate from the full-model profiler totals.

IV. Experimental Setup

A. Dataset

The dataset was collected from a real circular-knitting production line and covers three defect categories: horizontal defects (*row*), vertical defects (*col*), and holes (*hole*). It comprises 403 images with 468 annotated defect instances, divided into training, validation, and test sets of 282, 40, and 81 images, respectively, as summarized in Table II.

TABLE II
IMAGE-LEVEL SPLIT AND DEFECT-INSTANCE COUNTS IN THE REAL PRODUCTION-LINE DATASET.

Split	Images	Instances	Row	Col	Hole
Train	282	329	174	75	80
Validation	40	48	24	13	11
Test	81	91	60	12	19

B. Implementation Details

TABLE III
IMPLEMENTATION PROTOCOL COVERING TRAINING (ULTRALYTICS) AND DEPLOYMENT (EDGE DEVICE).

Category	Parameter	Specification
Training setup	Hardware	Intel i7-13700KF; NVIDIA RTX 3090 (24 GB)
	Software	Ubuntu 24.04, Python 3.10.20, PyTorch 2.4.1+cu124, and Ultralytics 8.4.140
Deployment setup	Hardware	Rockchip RK3566
	System	Android 13
	Quantization	INT8
Data protocol	Input resolution	Fixed at 640×640
Training dynamics	Batch	16
	Maximum epochs	300
	Patience	80 epochs
	Optimizer	AdamW

Table III summarizes the training and deployment settings. MCGI-YOLO and its component-ablation variants share the same `yolov8n.pt` initialization as the YOLOv8n baseline, and all component comparisons use a common training protocol with matched random seeds. Other comparison models use available pretrained checkpoints from their respective implementations.

C. Evaluation Metrics

Intersection over union (IoU) measures the overlap between predicted and ground-truth bounding boxes as the ratio of their intersection area to their union area. Detection accuracy is evaluated using mAP50 and mAP50–95. Let $a_c(t)$ denote the average precision for class c at IoU threshold t. The metrics are computed as

$$\mathrm{mAP}_{50} = \frac{1}{N_c} \sum_{c=1}^{N_c} a_c(0.50) \tag{19}$$

$$\mathrm{mAP}_{50:95} = \frac{1}{10 N_c} \sum_{c=1}^{N_c} \sum_{t \in T} a_c(t) \tag{20}$$

where $N_c = 3$ and $T = \{0.50, 0.55, \ldots, 0.95\}$. Both metrics are reported as percentages, with improvements expressed in percentage points.

Model parameters and FLOPs are measured at 640×640 input resolution using each model's respective framework. For FPS comparison, all models are evaluated on an RTX 3090 with FP32 precision, batch size 1, and 640×640 inputs. This measurement is separate from the RK3566 deployment result.

V. Results and Discussion

A. Comparison With Other Detectors

Table IV compares test-set accuracy, model complexity, and inference throughput. MCGI-YOLO achieves the highest mAP50 of 97.4% and the second-highest mAP50–95 of 65.5%, improving over the YOLOv8n [27] baseline by 6.9 and 7.1 percentage points, respectively. This gain comes with only a marginal increase in parameters (3.01 M $\rightarrow$ 3.02 M) and FLOPs (8.1 G $\rightarrow$ 8.2 G), while throughput on the RTX 3090 decreases from 554 to 491 FPS. Among lightweight variants, YOLO11n [27] attains the next-best accuracy (93.3% mAP50, 60.8% mAP50–95); MCGI-YOLO surpasses it by 4.1 and 4.7 points while running faster (491 vs. 463 FPS). YOLOv9t [30], YOLOv10n [31], YOLO12n [32], and YOLO26n [27] have fewer parameters or lower FLOPs but underperform on this test set, confirming that task-specific adaptation outperforms generic lightweight detectors. RT-DETR-R18 [33] leads in mAP50–95 (67.7%) but requires 20.09 M parameters and 60.7 G FLOPs at 146 FPS. MCGI-YOLO also outperforms DETR-R50 [34], Deformable DETR-R50 [35], and Faster R-CNN-R50 [36] on both mAP metrics with substantially lower cost, offering the best overall accuracy–efficiency tradeoff. Fig. 6 shows that MCGI-YOLO produces fewer fragmented boxes and tighter localization on elongated defects and holes, consistent with the aggregate results.

B. Component Ablation

Table V compares the baseline, SGRE alone, SARU alone, and their combination. The SGRE-only model retains nearest-neighbor upsampling; the SARU-only model retains the original backbone. SGRE alone achieves 94.3% mAP50 and 61.5% mAP50–95, improving over the baseline by 3.8 and 3.1 percentage points. SARU alone achieves 93.4% and 61.6%, corresponding to gains of 2.9 and 3.2 points. Their combination achieves 97.4% and 65.5%, exceeding the best single-module result by 3.1 and 3.9 points for the two metrics. These results indicate complementary benefits from progressive gradient-based refinement and subpixel-adaptive residual reconstruction.

TABLE IV
DETECTOR COMPARISON ON THE TEST SET. FPS DENOTES FP32 MODEL-FORWARD THROUGHPUT ON AN RTX 3090 WITH BATCH SIZE 1 AND A 640 × 640 INPUT. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY.

Model	Params (M)	FLOPs (G)	mAP50 (%)	mAP50–95 (%)	FPS
YOLOv9t [30]	**2.12**	8.5	90.1	53.3	291
YOLOv10n [31]	2.78	8.8	88.2	58.5	<u>524</u>
YOLO11n [27]	2.62	<u>6.7</u>	93.3	60.8	463
YOLO12n [32]	2.60	7.6	89.0	56.8	326
YOLO26n [27]	<u>2.57</u>	**6.2**	90.5	58.2	425
RT-DETR-R18 [33]	20.09	60.7	95.2	**67.7**	146
DETR-R50 [34]	41.28	77.3	96.4	53.6	101
Deformable DETR-R50 [35]	39.8	158.7	<u>96.8</u>	64.3	49
Faster R-CNN-R50 [36]	41.1	181.5	96.1	64.7	66
YOLOv8n (Baseline) [27]	3.01	8.1	90.5	58.4	**554**
MCGI-YOLO	3.02	8.2	**97.4**	<u>65.5</u>	491

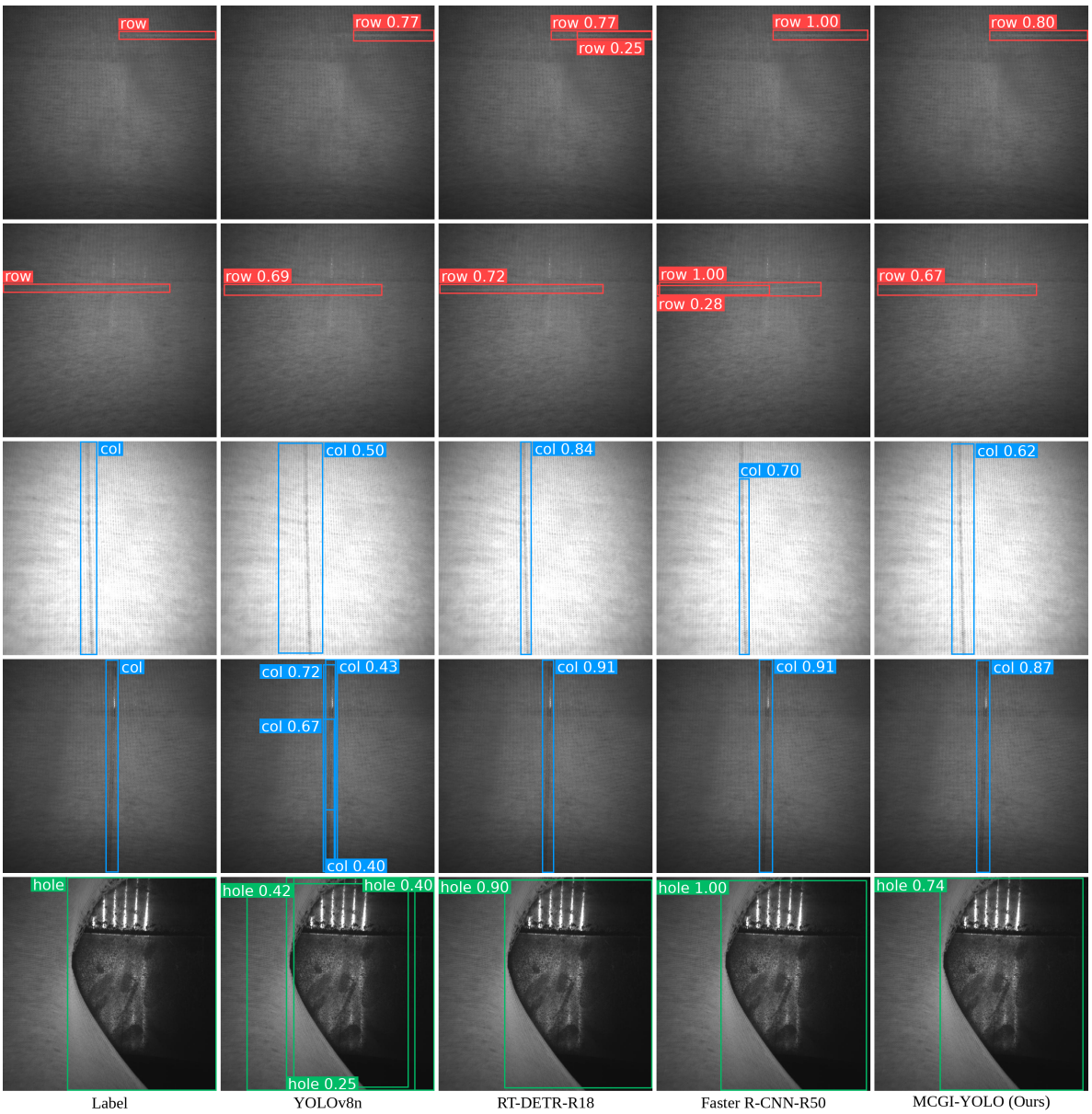


Fig. 6. Qualitative comparison of detection results across different methods. From left to right: ground-truth annotations (Label), YOLOv8n, RT-DETR-R18, Faster R-CNN-R50, and MCGI-YOLO (Ours).

TABLE V
COMPONENT ABLATION OF SGRE AND SARU.

Variant	SGRE	SARU	Params (M)	FLOPs (G)	mAP50	mAP50–95
YOLOv8n	–	–	3.01	8.1	90.5	58.4
+ SGRE	✓	–	3.02	8.2	94.3	61.5
+ SARU	–	✓	3.01	8.1	93.4	61.6
MCGI-YOLO	✓	✓	3.02	8.2	97.4	65.5

C. Model Export

The trained detector is exported to ONNX with a fixed input size of 640×640. The export retains the fixed Sobel filters, learned gradient-selection branches, and the PixelShuffle-based subpixel weighting operation. The ONNX model can then be converted to RKNN using RKNN-Toolkit2 [37] for the selected Rockchip target platform.

VI. CONCLUSION

This paper presents a dedicated defect detection and closed-loop control system for circular knitting machines, integrating image acquisition, SerDes transmission, edge inference, machine stopping, and cloud alerts. The lightweight MCGI-YOLO combines SGRE for progressive gradient-based feature enhancement with SARU for subpixel-adaptive residual reconstruction. On the self-collected test set, it achieves 97.4% mAP50 and 65.5% mAP50–95, improving over YOLOv8n by 6.9 and 7.1 percentage points. The detector reaches 491 FPS in pure FP32 inference on an RTX 3090 with batch size 1. Unlike generic object detection pipelines, our work jointly tailors both the hardware architecture and the detection algorithm to the morphology of knitting defects and the constraints of edge deployment, enabling real-time closed-loop control directly on the machine. By delivering a fully integrated and deployable system rather than an isolated algorithm, this work addresses a critical bottleneck in textile manufacturing—namely, the difficulty of translating detection performance into actionable, on-machine quality control. It offers the industry a ready-to-deploy reference for intelligent detection of circular knitting machines and, more broadly, demonstrates a reproducible pathway for deploying lightweight vision models on resource-constrained edge devices in industrial settings.

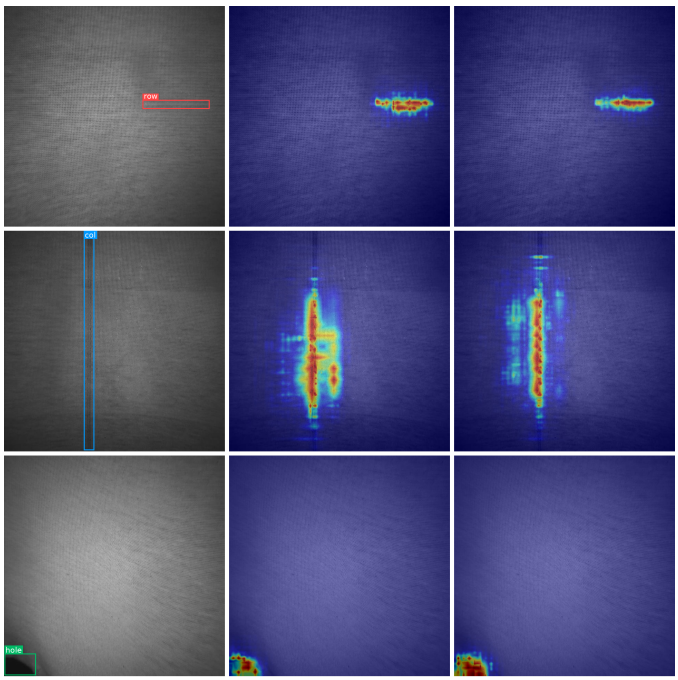


Fig. 7. Feature visualization of YOLOv8n and MCGI-YOLO. From left to right: ground-truth annotations (Label), YOLOv8n, and MCGI-YOLO.

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